

DSA0503 - Query Processing for Data Science

Experiments 21-30

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EXPERIMENT 21

Question: Write a Pandas program to swap the cases of a specified character column in a given DataFrame.

Program / Code

```
import pandas as pd

df = pd.DataFrame({
    "name": ["Alberto Franco", "Gino Mcneill",
            "Ryan Parkes", "Eesha Hinton",
            "David Parkes"]
})

df["name"] = df["name"].str.swapcase()

print(df)
```

Output

	name
0	aLBERTO fRANCO
1	gINO mCNEILL
2	rYAN pARKES
3	eESHA hINTON
4	dAVID pARKES

EXPERIMENT 22

Question: Write a Python program to draw a line with suitable label in the x axis, y axis and a title.

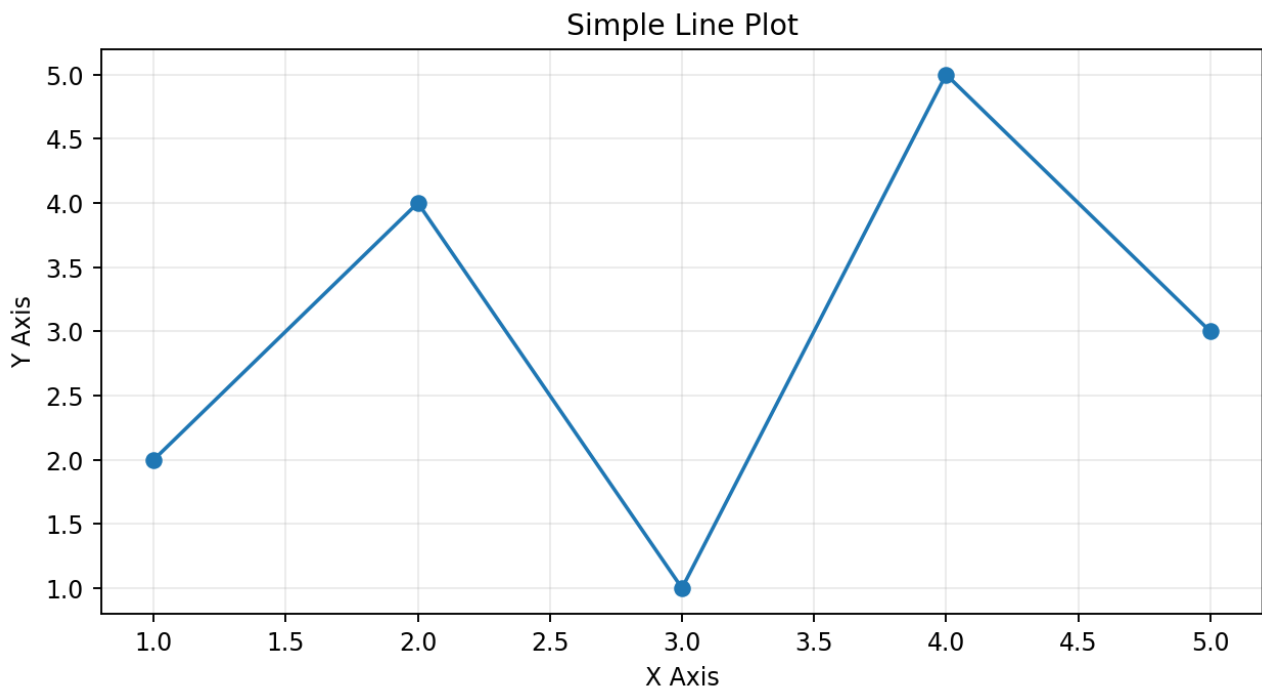
Program / Code

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [2, 4, 1, 5, 3]

plt.plot(x, y, marker="o")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Simple Line Plot")
plt.grid(True)
plt.show()
```

Output



EXPERIMENT 23

Question: Write a Python program to draw a line using given axis values taken from a text file, with suitable label in the x axis, y axis and a title.

Test Data:

test.txt

1 2

2 4

3 1

Program / Code

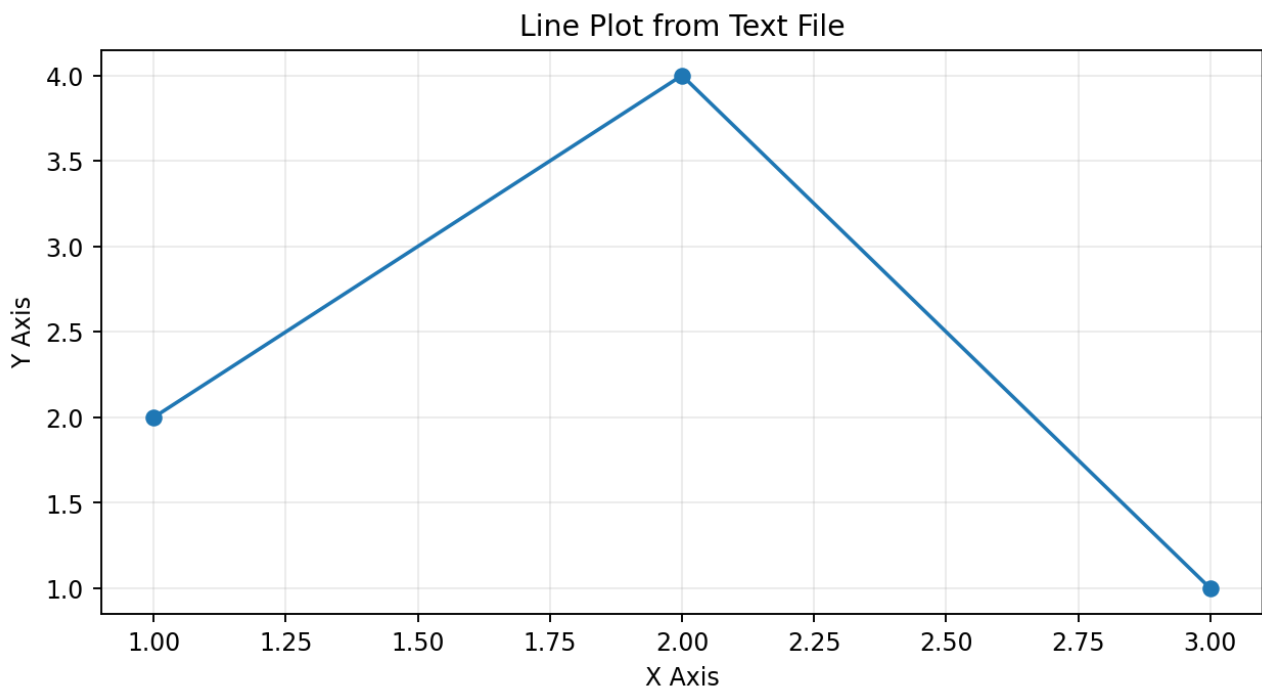
```
import matplotlib.pyplot as plt

x = []
y = []

with open("test.txt", "r") as file:
    for line in file:
        a, b = map(float, line.split())
        x.append(a)
        y.append(b)

plt.plot(x, y, marker="o")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Line Plot from Text File")
plt.grid(True)
plt.show()
```

Output



EXPERIMENT 24

Question: Write a Python program to draw line charts of the financial data of Alphabet Inc. between October 3, 2016 to October 7, 2016.

Sample Financial data (fdata.csv):

```
Date,Open,High,Low,Close
10-03-16,774.25,776.065002,769.5,772.559998
10-04-16,776.030029,778.710022,772.890015,776.429993
10-05-16,779.309998,782.070007,775.650024,776.469971
10-06-16,779.780.47998,775.539978,776.859985
10-07-16,779.659973,779.659973,770.75,775.080017
```

Program / Code

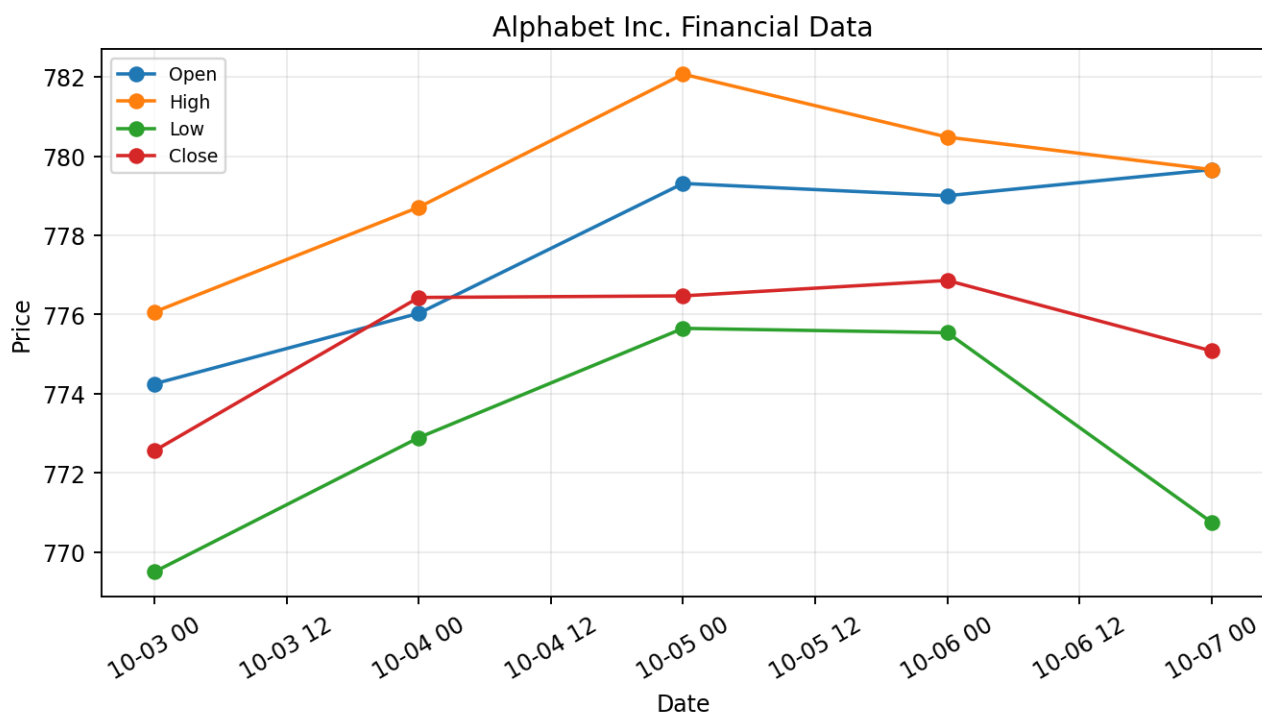
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("fdata.csv")
df["Date"] = pd.to_datetime(df["Date"])

plt.plot(df["Date"], df["Open"], marker="o", label="Open")
plt.plot(df["Date"], df["High"], marker="o", label="High")
plt.plot(df["Date"], df["Low"], marker="o", label="Low")
plt.plot(df["Date"], df["Close"], marker="o", label="Close")

plt.xlabel("Date")
plt.ylabel("Price")
plt.title("Alphabet Inc. Financial Data")
plt.legend()
plt.grid(True)
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```

Output



EXPERIMENT 25

Question: Write a Python program to plot two or more lines with legends, different widths and colors.

Program / Code

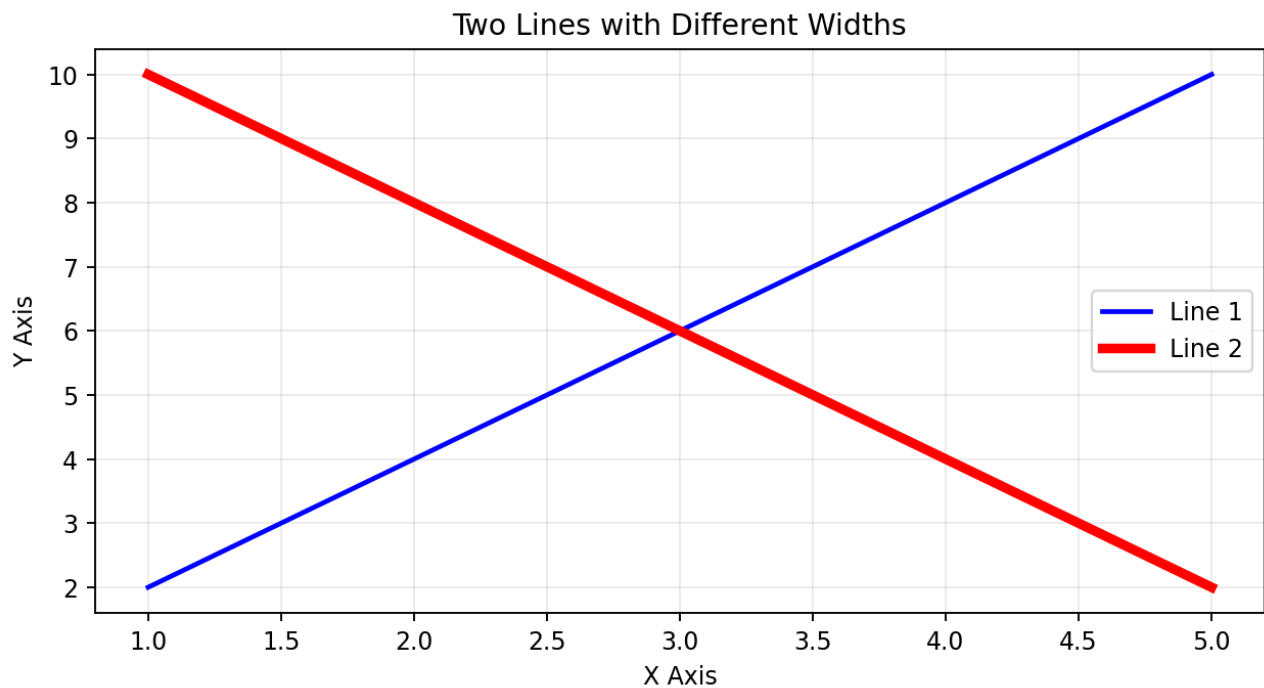
```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y1 = [2, 4, 6, 8, 10]
y2 = [10, 8, 6, 4, 2]

plt.plot(x, y1, color="blue", linewidth=2, label="Line 1")
plt.plot(x, y2, color="red", linewidth=4, label="Line 2")

plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Two Lines with Different Widths")
plt.legend()
plt.grid(True)
plt.show()
```

Output



EXPERIMENT 26

Question: Write a Python program to create multiple plots.

Program / Code

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]

plt.subplot(2, 2, 1)
plt.plot(x, [2, 4, 6, 8, 10])
plt.title("Line Plot")

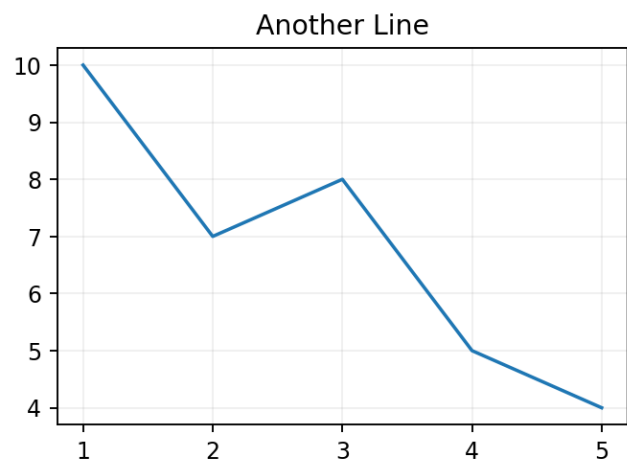
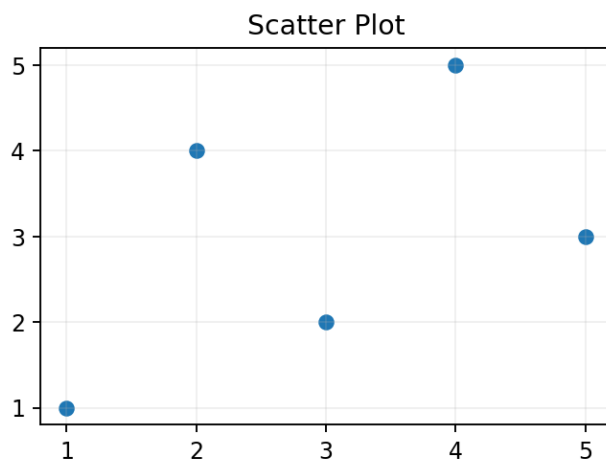
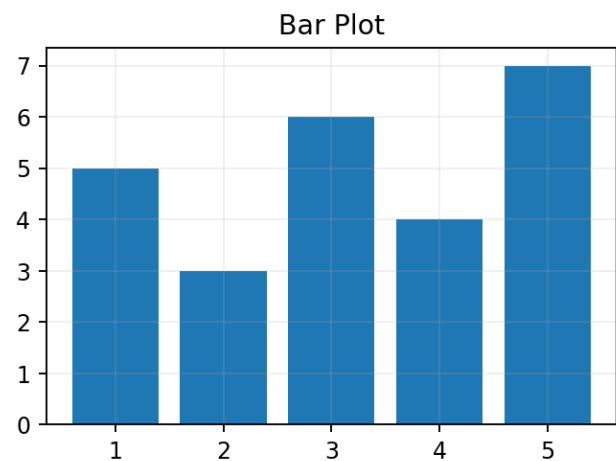
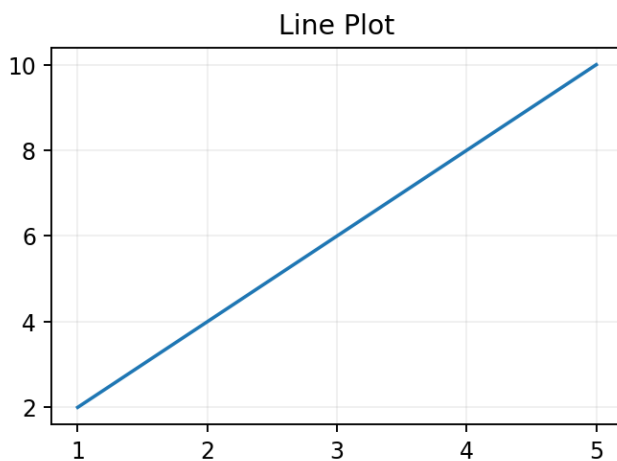
plt.subplot(2, 2, 2)
plt.bar(x, [5, 3, 6, 4, 7])
plt.title("Bar Plot")

plt.subplot(2, 2, 3)
plt.scatter(x, [1, 4, 2, 5, 3])
plt.title("Scatter Plot")

plt.subplot(2, 2, 4)
plt.plot(x, [10, 7, 8, 5, 4])
plt.title("Another Line")

plt.tight_layout()
plt.show()
```

Output



EXPERIMENT 27

Question: Write a Python programming to display a bar chart of the popularity of programming Languages.

Sample data:

Programming languages: Java, Python, PHP, JavaScript, C#, C++

Popularity: 22.2, 17.6, 8.8, 8, 7.7, 6.7

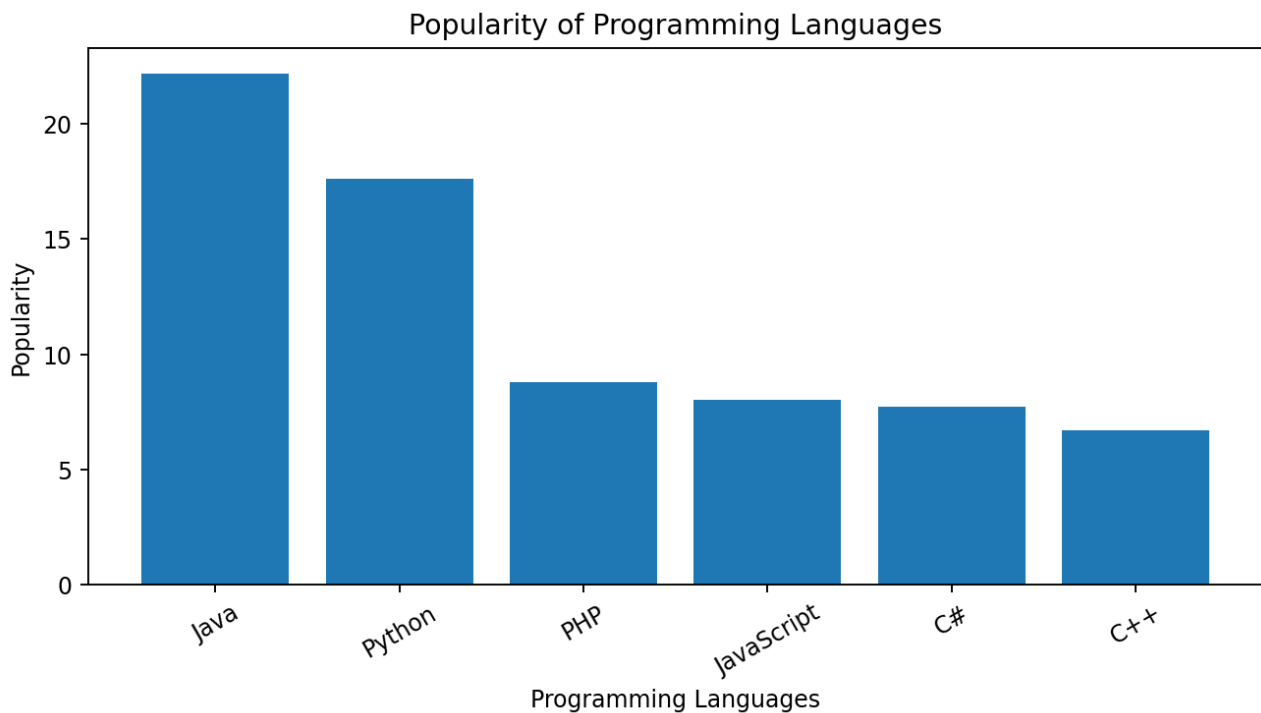
Program / Code

```
import matplotlib.pyplot as plt

languages = ["Java", "Python", "PHP", "JavaScript", "C#", "C++"]
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

plt.bar(languages, popularity)
plt.xlabel("Programming Languages")
plt.ylabel("Popularity")
plt.title("Popularity of Programming Languages")
plt.xticks(rotation=30)
plt.show()
```

Output



EXPERIMENT 28

Question: Write a Python programming to display a horizontal bar chart of the popularity of programming Languages.

Sample data:

Programming languages: Java, Python, PHP, JavaScript, C#, C++

Popularity: 22.2, 17.6, 8.8, 8, 7.7, 6.7

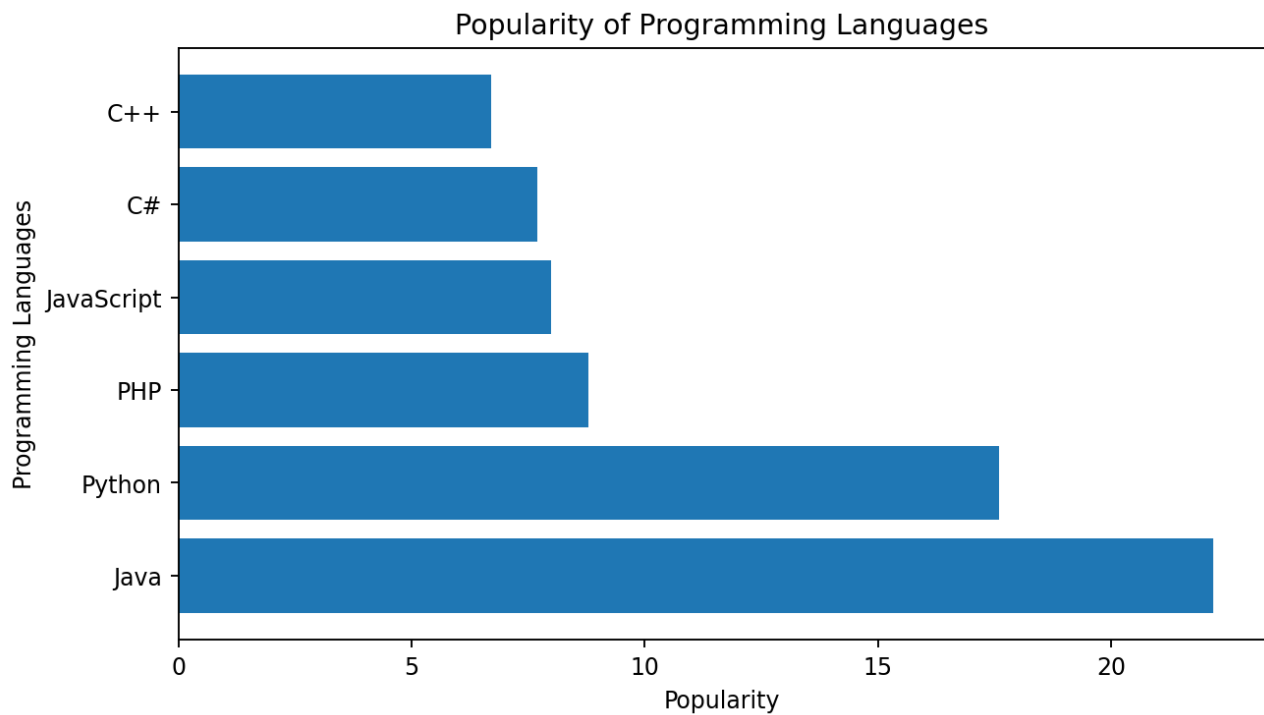
Program / Code

```
import matplotlib.pyplot as plt

languages = ["Java", "Python", "PHP", "JavaScript", "C#", "C++"]
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

plt.barh(languages, popularity)
plt.xlabel("Popularity")
plt.ylabel("Programming Languages")
plt.title("Popularity of Programming Languages")
plt.show()
```

Output



EXPERIMENT 29

Question: Write a Python programming to display a bar chart of the popularity of programming Languages. Use different color for each bar. of the popularity of programming Languages.

Sample data:

Programming languages: Java, Python, PHP, JavaScript, C#, C++

Popularity: 22.2, 17.6, 8.8, 8, 7.7, 6.7

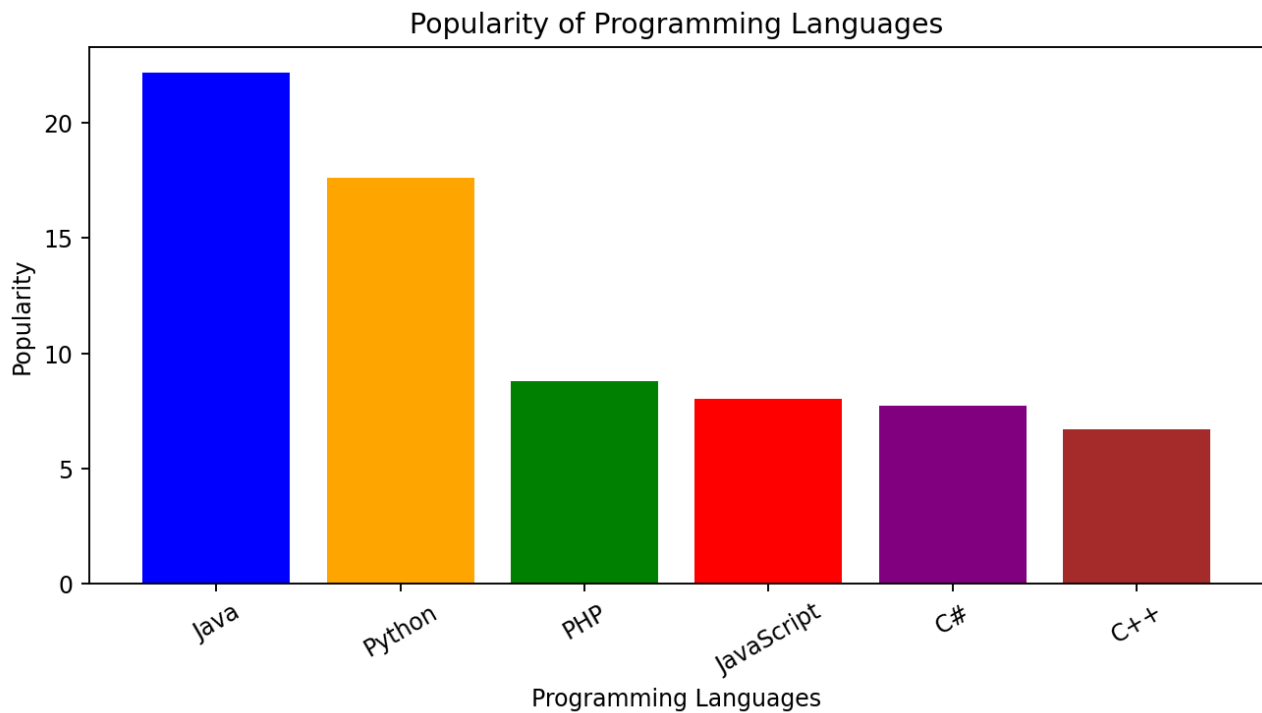
Program / Code

```
import matplotlib.pyplot as plt

languages = ["Java", "Python", "PHP", "JavaScript", "C#", "C++"]
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
colors = ["blue", "orange", "green", "red", "purple", "brown"]

plt.bar(languages, popularity, color=colors)
plt.xlabel("Programming Languages")
plt.ylabel("Popularity")
plt.title("Popularity of Programming Languages")
plt.xticks(rotation=30)
plt.show()
```

Output



EXPERIMENT 30

Question: Write a Python program to create bar plot of scores by group and gender. Use multiple X values on the same chart for men and women.

Sample Data:

Means (men) = (22, 30, 35, 35, 26)

Means (women) = (25, 32, 30, 35, 29)

Program / Code

```
import matplotlib.pyplot as plt
import numpy as np

groups = np.arange(5)
men = [22, 30, 35, 35, 26]
women = [25, 32, 30, 35, 29]

width = 0.35

plt.bar(groups - width/2, men, width, label="Men")
plt.bar(groups + width/2, women, width, label="Women")

plt.xlabel("Groups")
plt.ylabel("Scores")
plt.title("Scores by Group and Gender")
plt.xticks(groups, ["Group 1", "Group 2", "Group 3", "Group 4", "Group 5"])
plt.legend()
plt.show()
```

Output

